Quantum Seeds from the Repaired SFT Background

Phase, Interference, Detector Tagging, and Localization

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Abstract

This note records a first cluster of quantum-facing results suggested by the repaired effective action of Solvency Field Theory (SFT). The goal is deliberately narrow. We do *not* attempt a replacement derivation of quantum mechanics, nor do we claim the Born rule, operator algebra, Bell correlations, spin-statistics, or exclusion. Instead we isolate three seed results that can be established cleanly on top of the repaired SFT background. First, the standard matter WKB phase gives a Compton clock in proper time, whose Lorentz-transformed lab-frame phase yields the de Broglie relation and the ordinary two-path fringe law. Second, detector path-tagging suppresses interference exactly by reducing the overlap of path-conditioned detector states; the reduced path density matrix makes this transparent. Third, localization broadens momentum support: a Gaussian packet saturates the Fourier uncertainty bound $\Delta x\,\Delta p = \hbar/2$, and a finite slit of width a produces a transverse momentum scale $p_\perp \sim h/a$. The point of the note is to preserve the bridge cleanly and mathematically: the repaired SFT background is compatible with a "particle as ticking clock" interpretation of matter phase, with interference as phase-difference accumulation and path-tagging as loss of coherence through detector-record overlap.

1 Scope and Starting Point

The repaired effective action of SFT leaves the matter sector standard. The working action is

$$S_{\rm SFT}^{\rm eff} = \int d^4x\,\sqrt{-g}\,\left[\frac{R}{16\pi G} - \eta\,\nabla_\mu(\bar\chi+\psi)J^\mu_{\rm ops} + \Lambda_C\,\mathcal{C}_{\rm min} - V_0\left(e^{-\psi}-1\right)^2 + \mathcal{L}_m\right]. \tag{1}$$

Here $\bar\chi$ is the cosmological bookkeeping branch, ψ is the local lag relative to that branch, and $\mathcal{L}_m$ is the ordinary matter Lagrangian. The present note uses only one consequence of this structure: standard matter can be studied on the repaired SFT background without modifying the matter equations at leading WKB order.

The targets of this note are therefore intentionally modest:

1. show that the WKB phase of massive matter yields a Compton clock and, in a moving frame, the de Broglie wavelength;
2. show that two-path interference is controlled by phase difference and that detector-induced path records suppress the interference term by the detector-state overlap;

3. show that localization of spatial support broadens momentum support, both for Gaussian packets and for finite slit apertures.

The note should be read as a first bridge document. It is not a theorem that SFT derives the full quantum formalism. It is a precise record of the first three quantum-facing bridges that the repaired SFT background already supports.

2 Seed 1: Compton Phase, de Broglie Wavelength, and Two-Path Interference

2.1 WKB phase of a massive particle

Take a standard scalar matter field in WKB form,

$$\Psi(x) = A(x)e^{iS(x)/\hbar}. \tag{2}$$

At leading order, the phase S obeys the Hamilton–Jacobi equation. Along a timelike worldline of a massive particle, the on-shell action may be written as

$$S = -mc^2\tau, \tag{3}$$

where τ is proper time.

The phase is therefore

$$\theta = \frac{S}{\hbar} = -\frac{mc^2}{\hbar}\tau. \tag{4}$$

Differentiating with respect to proper time gives

$$\frac{d\theta}{d\tau} = -\frac{mc^2}{\hbar}. \tag{5}$$

Hence the intrinsic proper-time angular frequency is

$$\boxed{\omega_C = \frac{mc^2}{\hbar}.} \tag{6}$$

This is the Compton frequency. In the language of the repaired SFT background, this is the clean mathematical version of "the particle ticks at its Compton rate."

2.2 Lorentz transformation to the lab frame

Let the particle move in the x-direction with speed v. In its rest frame the phase is

$$\theta = -\omega_C t'. \tag{7}$$

Lorentz transformation gives

$$t' = \gamma\left(t - \frac{vx}{c^2}\right), \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \tag{8}$$

Therefore

$$\theta = -\frac{mc^2}{\hbar}\gamma\left(t - \frac{vx}{c^2}\right) \tag{9}$$

$$= \frac{-\gamma mc^2 t + \gamma mvx}{\hbar}. \tag{10}$$

Defining the usual energy and momentum,

$$E = \gamma mc^2, \qquad p = \gamma mv, \tag{11}$$

we obtain the lab-frame phase

$$\boxed{\theta = \frac{-Et + px}{\hbar}.} \tag{12}$$

Thus the moving phase is nothing more than the Lorentz-transformed Compton clock.

2.3 de Broglie wavelength

At fixed time, a full spatial phase cycle satisfies

$$\Delta\theta = \frac{p\,\Delta x}{\hbar} = 2\pi. \tag{13}$$

Hence the spatial period is

$$\lambda = \frac{2\pi\hbar}{p} = \frac{h}{p}. \tag{14}$$

So the de Broglie relation follows immediately:

$$\boxed{\lambda = \frac{h}{p}.} \tag{15}$$

This is the first bridge statement of the note: the proper-time face of the massive phase is the Compton clock, while the spatial face of the same phase in a moving frame is the de Broglie wavelength.

2.4 Two-path interference and the double slit

Suppose the particle can propagate to a screen along two paths γ_1 and γ_2. The phase difference is

$$\Delta\theta = \frac{1}{\hbar}(S_1 - S_2). \tag{16}$$

In the stationary equal-energy approximation, this reduces to

$$\Delta\theta \approx \frac{p\,\Delta L}{\hbar}, \tag{17}$$

where ΔL is the path-length difference.

Constructive interference requires

$$\Delta\theta = 2\pi n, \tag{18}$$

so

$$\Delta L = n\frac{2\pi\hbar}{p} = n\lambda. \tag{19}$$

For a double slit with slit separation d and screen angle ϑ, one has

$$\Delta L \approx d\sin\vartheta, \tag{20}$$

whence the ordinary fringe law

$$\boxed{d\sin\vartheta = n\lambda.} \tag{21}$$

For small angles, $\sin\vartheta \approx y/L$, so the fringe positions are

$$\boxed{y_n = \frac{n\lambda L}{d}.} \tag{22}$$

The bridge is therefore

$$\boxed{\text{Compton phase } \Rightarrow \text{ de Broglie wavelength } \Rightarrow \text{ ordinary two-path interference.}} \tag{23}$$

3 Seed 2: Detector Tagging and Suppression of Interference

3.1 Two-path state without a detector

With two path amplitudes A_1 and A_2, the state is

$$|\Psi\rangle = A_1\,|1\rangle + A_2\,|2\rangle\,. \tag{24}$$

At the screen, the amplitude is

$$\psi(x) = A_1\psi_1(x) + A_2\psi_2(x). \tag{25}$$

The corresponding intensity is

$$I(x) = |A_1\psi_1|^2 + |A_2\psi_2|^2 + 2\,\mathrm{Re}\,(A_1^* A_2 \psi_1^* \psi_2)\,. \tag{26}$$

The last term is the interference term.

3.2 Detector entanglement and reduced path state

Now include a detector that records which path was taken. The joint state becomes

$$|\Psi\rangle = A_1 |1\rangle |D_1\rangle + A_2 |2\rangle |D_2\rangle . \tag{27}$$

The full density matrix is

$$\rho = |\Psi\rangle \langle\Psi| . \tag{28}$$

Tracing over the detector yields the reduced density matrix on path space:

$$\boxed{\rho_{\text{path}} = |A_1|^2 |1\rangle \langle 1| + |A_2|^2 |2\rangle \langle 2| + A_1 A_2^* \gamma |1\rangle \langle 2| + A_1^* A_2 \gamma^* |2\rangle \langle 1| ,} \tag{29}$$

with

$$\boxed{\gamma = \langle D_2|D_1|D_2|D_1\rangle .} \tag{30}$$

Thus the detector does one exact thing: it multiplies the off-diagonal coherence by the overlap of the detector states.

3.3 Interference with detector overlap

For equal-amplitude paths, $A_1 = A_2 = 1/\sqrt{2}$, and path phases $\psi_1 = e^{i\phi/2}$, $\psi_2 = e^{-i\phi/2}$, the intensity becomes

$$I(\phi) = 2 + \gamma e^{-i\phi} + \gamma^* e^{i\phi}. \tag{31}$$

Writing

$$\gamma = |\gamma| e^{i\delta}, \tag{32}$$

this becomes

$$\boxed{I(\phi) = 2 + 2|\gamma| \cos(\phi - \delta).} \tag{33}$$

Hence the fringe visibility is

$$\boxed{\mathcal{V} = |\gamma|.} \tag{34}$$

Three limiting cases are immediate:

$$|\gamma| = 1 \quad \Rightarrow \quad \text{full coherence and full interference,} \tag{35}$$

$$0 < |\gamma| < 1 \quad \Rightarrow \quad \text{partial path information and partial visibility,} \tag{36}$$

$$|\gamma| = 0 \quad \Rightarrow \quad \text{perfect path tagging and no interference.} \tag{37}$$

3.4 Visibility, distinguishability, and purity

In the equal-amplitude case the reduced density matrix is

$$\rho_{\text{path}} = \frac{1}{2} \begin{pmatrix} 1 & \gamma \\ \gamma^* & 1 \end{pmatrix} . \tag{38}$$

Then the standard distinguishability measure is

$$\boxed{\mathcal{D} = \sqrt{1 - |\gamma|^2},} \tag{39}$$

so

$$\boxed{\mathcal{V}^2 + \mathcal{D}^2 = 1.} \tag{40}$$

The purity is

$$\boxed{\mathrm{Tr}(\rho_{\mathrm{path}}^2) = \frac{1}{2}(1 + |\gamma|^2).} \tag{41}$$

Thus

$$|\gamma| = 1 \quad \Rightarrow \quad \text{pure coherent superposition,} \tag{42}$$

$$|\gamma| = 0 \quad \Rightarrow \quad \text{maximally mixed two-path state.} \tag{43}$$

In SFT language, the detector creates path-specific bookkeeping records $|D_1\rangle$ and $|D_2\rangle$. Interference survives only to the extent that those records remain the same record, i.e.

$$|\gamma| = |\,\langle D_2|D_1|D_2|D_1\rangle\,|. \tag{44}$$

Measurement is therefore not treated here as a new postulate. At this seed level, it is path-to-record correlation plus reduced overlap of those records.

4 Seed 3: Localization, Momentum Spread, and Slit Diffraction

4.1 Gaussian packet and the Fourier bound

Take the normalized Gaussian wavefunction

$$\psi(x) = \left(\frac{1}{2\pi\sigma^2}\right)^{1/4} \exp\left(-\frac{x^2}{4\sigma^2}\right). \tag{45}$$

Then

$$\langle x\rangle = 0, \qquad \langle x^2\rangle = \sigma^2, \qquad \Delta x = \sigma. \tag{46}$$

Its Fourier transform is again Gaussian,

$$\phi(k) = \left(\frac{2\sigma^2}{\pi}\right)^{1/4} e^{-\sigma^2 k^2}, \tag{47}$$

with

$$\langle k\rangle = 0, \qquad \langle k^2\rangle = \frac{1}{4\sigma^2}, \qquad \Delta k = \frac{1}{2\sigma}. \tag{48}$$

Therefore

$$\boxed{\Delta x\,\Delta k = \frac{1}{2},} \qquad \boxed{\Delta x\,\Delta p = \frac{\hbar}{2}.} \tag{49}$$

So the localization–momentum-spread tradeoff appears exactly in the cleanest baseline case. A narrower packet in x necessarily carries broader momentum support.

4.2 Finite slit aperture

Now take a top-hat aperture of width a:

$$\psi_{\text{slit}}(x) = \begin{cases} \dfrac{1}{\sqrt{a}}, & |x| \le a/2, \\ 0, & \text{otherwise.} \end{cases} \tag{50}$$

Its far-field Fourier amplitude is

$$\phi_{\text{slit}}(k) = \sqrt{\frac{2}{\pi a}}\,\frac{\sin(ak/2)}{k}, \tag{51}$$

so the intensity envelope is

$$I_{\text{slit}}(k) = \frac{2}{\pi a}\frac{\sin^2(ak/2)}{k^2}. \tag{52}$$

This is the usual sinc diffraction pattern. The first zero occurs when

$$\frac{ak}{2} = \pi, \tag{53}$$

so

$$\boxed{k_{\text{first zero}} = \frac{2\pi}{a}.} \tag{54}$$

Multiplying by $\hbar$ gives the transverse momentum scale

$$\boxed{p_{\perp,\text{first zero}} = \hbar k_{\text{first zero}} = \frac{h}{a}.} \tag{55}$$

Thus a narrower slit forces a broader transverse momentum spread.

4.3 Interpretive bridge

The first uncertainty bridge can therefore be stated as follows:

$$\boxed{\text{narrowing spatial support} \;\Rightarrow\; \text{broadening momentum support.}} \tag{56}$$

For the Gaussian packet, the Fourier bound is saturated at $\hbar/2$. For the slit, the first diffraction scale is $p_\perp \sim h/a$. In SFT language, restricting the allowed bookkeeping support of the outgoing history sharpens the spatial write region but broadens the momentum and phase content of the outgoing state.

5 Status of the Three Seeds

The results of this note can be summarized compactly.

Established in this note

1. The WKB phase of a massive particle yields a proper-time Compton clock,

$$\omega_C = \frac{mc^2}{\hbar}. \tag{57}$$

2. Lorentz transformation of that same phase yields the de Broglie relation,

$$\lambda = \frac{h}{p}. \tag{58}$$

3. Two-path phase difference reproduces the ordinary fringe law,

$$d \sin \vartheta = n\lambda. \tag{59}$$

4. Detector path-tagging suppresses the interference term by the detector-state overlap,

$$\mathcal{V} = |\gamma|, \qquad \gamma = \langle D_2|D_1|D_2|D_1\rangle\,. \tag{60}$$

5. Localization broadens momentum support. In the Gaussian baseline,

$$\Delta x\, \Delta p = \frac{\hbar}{2}. \tag{61}$$

   In the slit baseline,

$$p_\perp \sim \frac{h}{a}. \tag{62}$$

Not established in this note

This note does *not* derive or claim:

1. the Born rule,
2. Hilbert-space structure from the SFT action itself,
3. canonical commutators or a derivation of the Heisenberg theorem from SFT,
4. Bell correlations or nonlocality constraints,
5. spin-statistics or Pauli exclusion.

6 Conclusion

The repaired SFT background supports a first nontrivial quantum bridge. Standard matter phase on that background already yields a clean Compton–de Broglie–interference chain. Detector path-tagging is naturally encoded as suppression of off-diagonal coherence by detector-record overlap, and spatial localization inevitably broadens momentum support in the ordinary Fourier sense. The content of the note is therefore narrow but real: it does not derive all of quantum mechanics, but it does show that the repaired SFT background is compatible with the first phase-based structures that any quantum bridge would have to reproduce.

The next honest targets are correspondingly clear: sharpen detector-tagging into a more explicit decoherence model, test whether SFT lag geometry perturbs phase accumulation in a controlled way, and only then move on to deeper targets such as uncertainty commutators, Bell structure, or spin-statistics.